Trainings

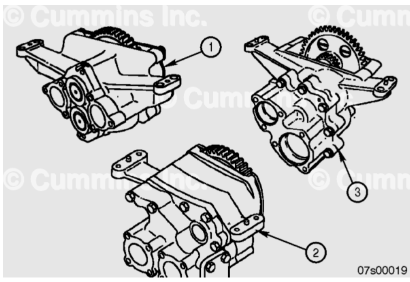
General Information

⚠ CAUTION ⚠

There are four different type of lubricating oil pumps used on K38. K50, QSK38 and QSK50 engines. If the lubricating oil pump is replaced, then the correct replacement pump MUST be installed.

All pumps are mounted to the bottom at the rear of the cylinder block. The pump is driven by a gear on the rear of the crankshaft. The main drive gear of the pump to the crankshaft gear backlash is critical for the durability of the pump and gears. The backlash is adjustable with shims between the pump mounting legs and the cylinder block. The shims are identical for K38, K50, QSK38, and QSK50 engines.

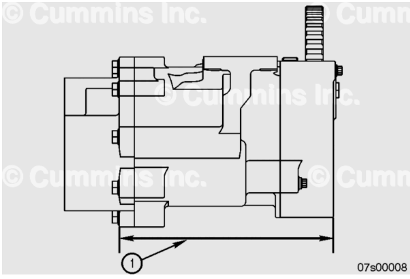
The lubricating oil pumps used on different engines are detailed in the table below.



Lubricating Oil Pumps	
Lubricating Oil Pump Type	**Where Used**
Two-Piece Pump (1)	K38 Engines
Three-Piece Pump (2)	K50 G-Drive Engines
	K50 Marine Auxilary Drive Engines
Slow Speed Pump (Short Body) (3)	QSK38 Engines
	K38 Marine Engines
	K50 Marine Engines
Slow Speed Pump (Long Body) (3)	QSK50 Engines
	K50-DR (1200 rpm) Engines

The two different slow speed pumps can be identified by measuring the length of the body from the front to the rear faces.

Low Speed Lubricating Oil Pumps	
Lubricating Oil Pump Type	**Distance (1)**
Slow Speed Pump (Short Body)	247 mm [9.72 in]
Slow Speed Pump (Long Body)	269 mm [10.6 in]



Preparatory Steps

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, make sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

⚠ CAUTION ⚠

Use caution when draining oil or replacing filters that the oil is not spilled or drained into the bilge area. The oil and oil filters must be disposed of in accordance with local environmental regulations.

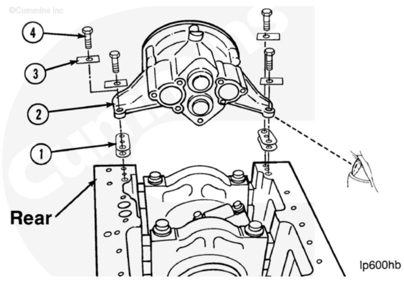
- Disconnect the batteries. Refer to the equipment manufacturer service information.
- Drain the lubricating oil. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-037-tr.html)

- Remove the full flow filters. Refer to Procedure 007-013 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-013-tr.html)
- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-025-tr.html)
- Remove the lubricating oil pan adapter. Refer to Procedure 007-027 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-027-tr.html)
- Remove the lubricating oil suction tube (block mounted). Refer to Procedure 007-035 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-035-tr.html)

Remove

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, make sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Note : Shown in this procedure is the K38. The procedure is the same for the K50, QSK38, KTA38GC, and QSK50 engines.

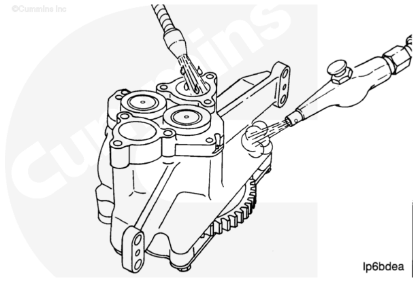
It is recommended that the oil pump-to-crankshaft gear lash be measured **before** removing the pump to check for worn gears. Instructions for measuring the gear lash are in the Install section of this procedure.

If lockplates are installed, bend the lockplates (3) from the heads of the capscrews and discard the lockplates. New lockplates **must** be installed when installing the pump.

Remove the four capscrews (4), washers, if installed, and the pump (2). Keep any shims (1) and record their original position.

Note : The dowel pins do not have to be removed unless they are damaged.

Clean and Inspect for Reuse



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturers recommendations for use. Wear goggles and protective clothing to reduce the risk of personal injury.

⚠ WARNING ⚠

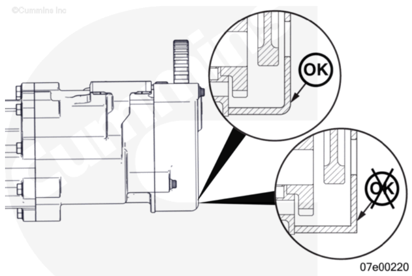
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use solvent to flush the interior of the pump thoroughly. Clean the exterior of the pump.

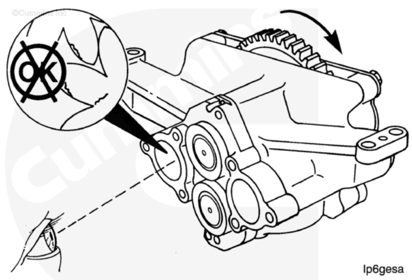
Dry with compressed air.

Note : This step only applies to the slow speed pumps.

Visually check the gear cover for the 0.44 inch [11.176mm] corner radius. If the gear cover does **not** have this radius the part **must** be replaced.

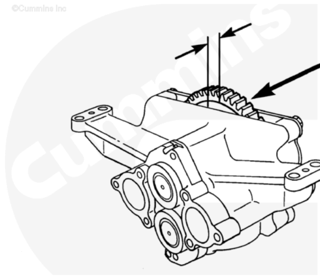


Use the main drive gear to rotate the pump. The pump **must** rotate freely. Check the drive and pumping gears. If any of the gears are damaged or the pump does **not** rotate freely, the pump **must** be replaced.



Move the main drive gear by hand when checking the end play of the main drive gear.

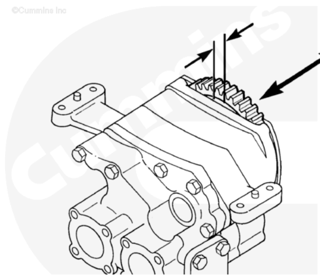
Two-Piece Pump and Three-Piece Pump Drive Gear End Play		
mm		in
0.10	MIN	0.004
0.18	MAX	0.007



lp6geja

If the end play is **not** within specifications, the part **must** be rebuilt.

Slow Speed Pump (Short Body and Long Body) Drive Gear End Play		
mm		in
0.20	MIN	0.008
0.81	MAX	0.032



lp6gejg

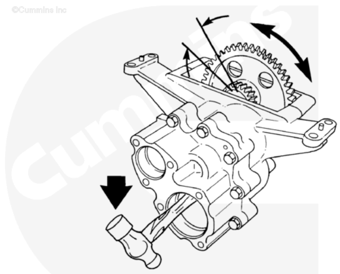
If the end play is **not** within specifications, the part **must** be rebuilt.

Note : This step only applies to slow speed pumps.

Note : Do not damage the pumping gears.

Use a piece of wood or the handle of a hammer to prevent the pumping gears from moving.

Move the gear by hand when checking the backlash.



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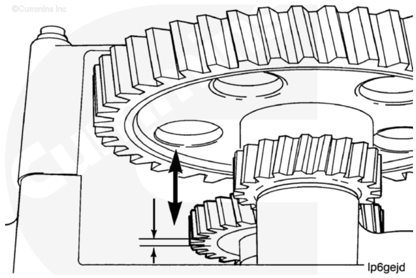
Slow Speed Pump (Short Body and Long Body) Compound Gear to Drive Gear Backlash		
mm		in
0.15	MIN	0.006
0.25	MAX	0.010

Note : This step only applies to slow speed pumps.

Move the gear by hand to check the clearance.



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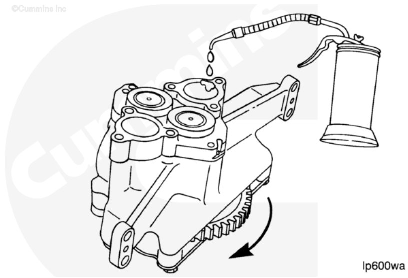
Slow Speed Pump (Short Body and Long Body) Drive Shaft End Play		
mm		in
0.114	MIN	0.0045
0.216	MAX	0.0085

If the end play is **not** within specifications, the part **must** be rebuilt.

Use clean engine oil. While turning the main drive gear, lubricate the pumping gears.

Cover the pump to prevent the entry of dirt until the pump is to be installed on the engine.



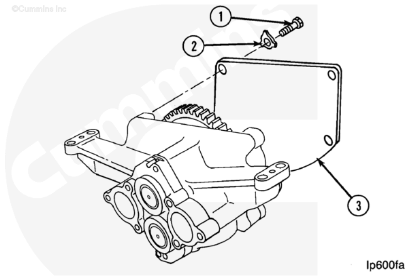
Disassemble

Two-Piece

Use a hammer and a drift to bend the lockplates from the capscrews.

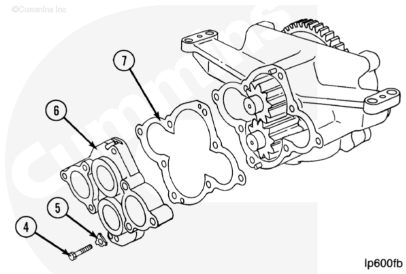
Remove the five capscrews (1) and the lockplates (2). Discard the lockplates.

Remove the main drive gear cover (3).



⚠ CAUTION ⚠

Do not damage the gasket surface when separating the cover from the body.



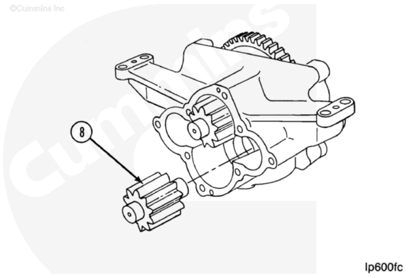
Use a hammer and a drift to bend the lockplates from the capscrews.

Remove the two capscrews (4) and the lockplates (5). Discard the lockplates.

Note : The holes for the dowel pins in the cover contain 5/16-18 threads. Use a capscrew in both holes to push the cover from the dowels.

Remove the cover (6) and gasket (7). Discard the gasket.

Remove the pumping idler gear and shaft assembly (8).

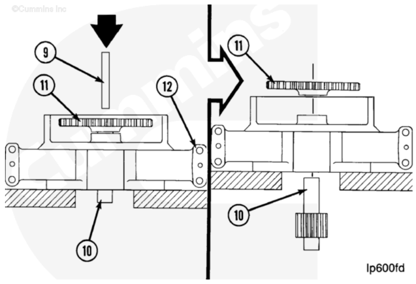


Use a hydraulic press and support the body (12) as illustrated. Make sure there is sufficient clearance for the pumping gear to be moved from the body.

Use a mandrel (9) that is smaller than the bore in the drive gear.

Remove the pumping drive gear and shaft assembly (10).

Remove the main drive gear (11).

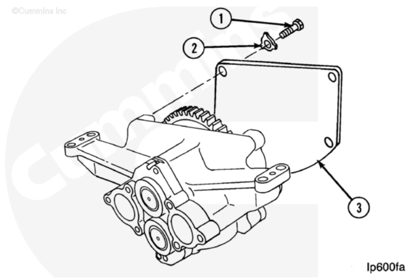


Three-Piece

Use a hammer and a drift to bend the lockplates from the capscrews.

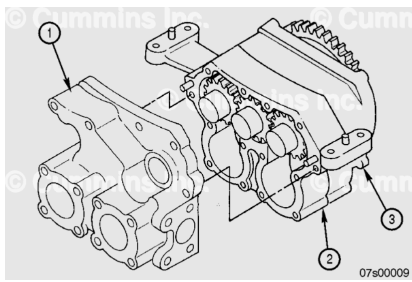
Remove the five capscrews (1) and the lockplates (2). Discard the lockplates.

Remove the main drive gear cover (3).

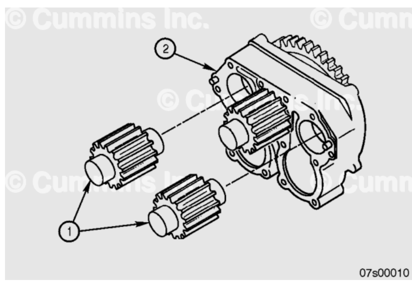


Remove the six capscrews from the front cover (1).

Separate the front cover (1), main body (2), and rear cover (3), taking care **not** to damage the mating surfaces.



Remove the two short pumping gears (1) and shaft assemblies from the pump (2).

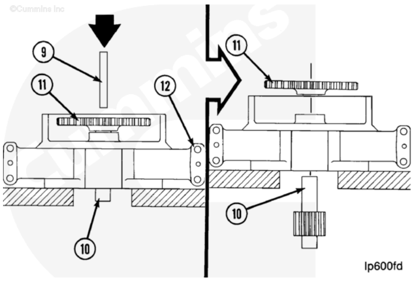


Use a hydraulic press and support the body (12) as illustrated. Make sure there is sufficient clearance for the pumping gear to be moved from the body.

Use a mandrel (9) that is smaller than the bore in the drive gear.

Remove the pumping drive gear and shaft assembly (10).

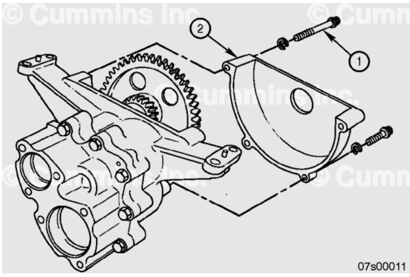
Remove the main drive gear (11).



Slow-Speed

Remove the four capscrews (1) from the drive gear cover (2).

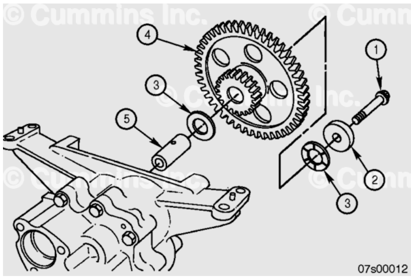
Remove the drive gear cover (2).



Remove the capscrew (1), retainer(2), and thrust washers (3) from the compound drive / idler gear (4).

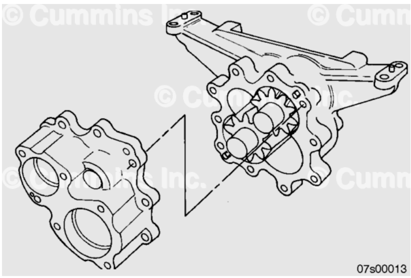
Remove the compound drive / idler gear (4).

Remove the idler shaft (5).

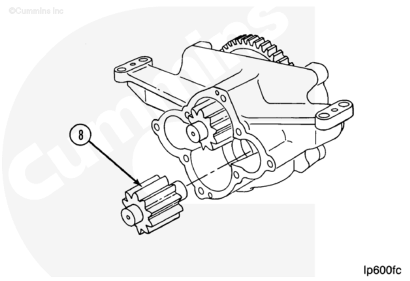


Remove the ten capscrews and washers from the front cover.

Remove the front cover, taking care not to damage the mating surfaces.



Remove the pumping idler gear and shaft assembly (8).

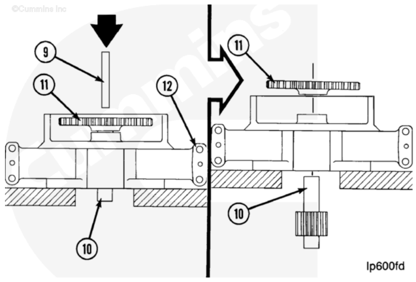


Use a hydraulic press and support the body (12) as illustrated. Make sure there is sufficient clearance for the pumping gear to be moved from the body.

Use a mandrel (9) that is smaller than the bore in the drive gear.

Remove the pumping drive gear and shaft assembly (10).

Remove the drive gear (11).



Clean for Rebuild

⚠ WARNING ⚠

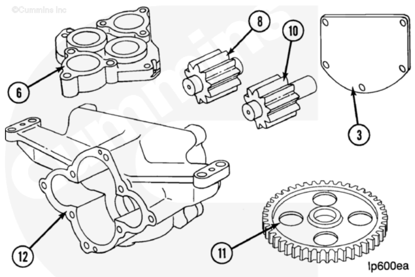
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturers recommendations for use. Wear goggles and protective clothing to reduce the risk of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use solvent to clean the parts.

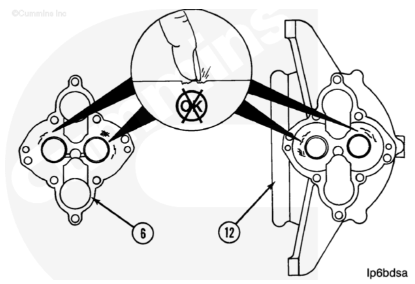
Dry with compressed air.



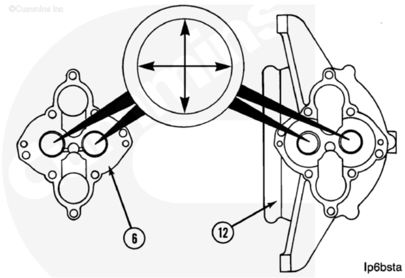
Inspect

Check the cover (6) and the body (12) for damage caused by the pumping gears.

If the marks on the cover or body can be felt with a fingernail, the part **must** be replaced.



Measure the inside diameter of the bushings in the cover (6) and body (12).

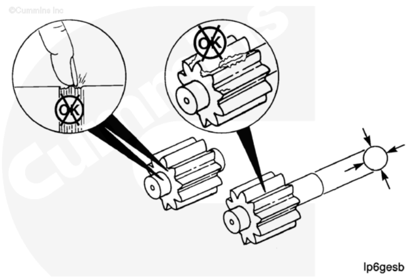


Oil Pump Bushing Inside Diameter		
mm		in
38.164	MIN	1.5025
38.227	MAX	1.5050

Note : If either of the bushings in the body or the cover are not within specifications, the assembly must be repaired or replaced.

Note : The illustration shows a two-piece pump, but the specifications of the bushings are the same for all types of lubricating oil pump.

Check the shaft and gear assemblies for damage or wear.



If the damage on the shaft can be felt with a fingernail, the assembly **must** be replaced.

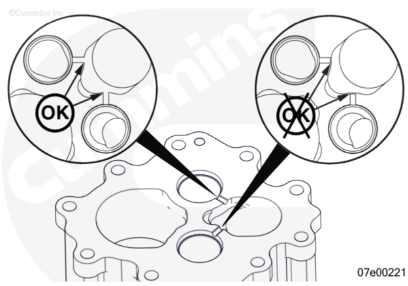
Measure the outside diameter of the oil pump shaft. Replace if out of specifications.

Oil Pump Shaft Outside Diameter		
mm		in
38.082	MIN	1.4993
38.100	MAX	1.5000

Check the gears for pitting at the roots of the gear teeth. Pits in this area are a sign of cavitation.

If the gear contains pits, cracks, or broken teeth, the gear and shaft assembly **must** be replaced.

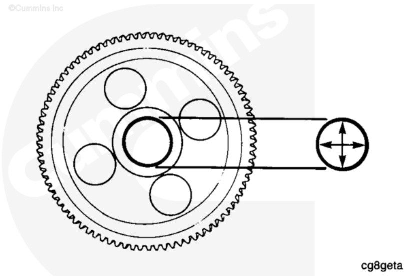
Note : This step only applies to the slow speed pump.



Check inside the pump body where the pumping gears were housed. Check the two oil transverse oil drilling to make sure they are complete (should break out into the smaller oil cavity.

For the two-piece and three-piece lubricating oil pumps, check the main drive gear for cracks or other damage. If the part is cracked or otherwise damaged, the part **must** be replaced.

Measure the inside diameter of the oil pump drive gear. Replace if out of specifications.



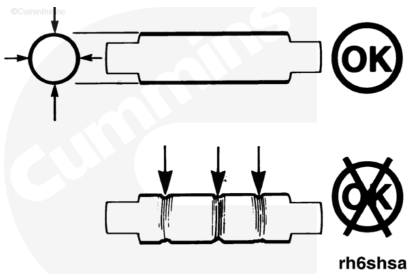
Two-Piece Pump and Three-Piece Pump Oil Pump Drive Gear Inside Diameter		
mm		in
38.031	MIN	1.4973
38.057	MAX	1.4983

For the slow speed lubricating oil pumps, check the compound drive / idler gear for cracks or other damage. If the part is cracked or otherwise damaged, the part must be replaced.

Measure the inside diameter of the compound drive / idler gear bushing. Replace if out of specifications.

Slow Speed Pump Compound Drive / Idler Gear Bushing Inside Diameter		
mm		in
25.413	MIN	1.0005
25.437	MAX	1.0015

Note : This step only applies to the slow speed lubricating oil pumps



Check the compound drive / idler gear shaft for cracks or other damage. If the part is cracked or otherwise damaged, the part must be replaced.

Measure the outside diameter of the compound drive / idler gear shaft. Replace if out of specifications.

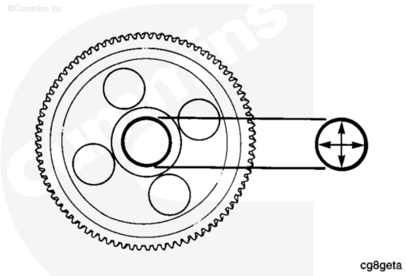
Slow Speed Pump Compound Drive / Idler Gear Shaft Outside Diameter		
mm		in
25.3619	MIN	0.9985
25.3746	MAX	0.9990

Note : This step only applies to the slow speed lubricating oil pumps

Check drive gear for cracks or other damage. If the part is cracked or otherwise damaged, the part must be replaced.

Measure the inside diameter of the drive gear. Replace the gear if out of specifications.

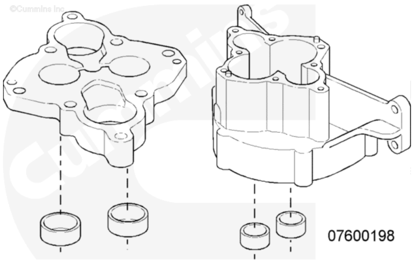
Slow Speed Pump Compound Drive / Idler Gear Bushing Inside Diameter		
mm		in
38.031	MIN	1.4973
38.057	MAX	1.4983



Repair

Remove all bushings that are to be replaced in both the body and cover.

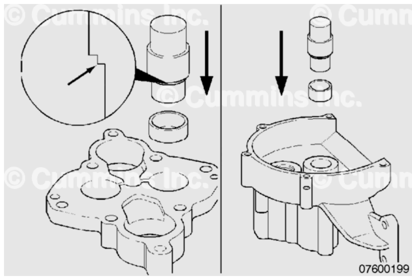
Clean and deburr the bores in the housing and cover.



Use driver, Part Number 3822289, and install the new bushings .

When there is an oil groove connecting with the bore, make sure the top of the bushing is flush or below the bottom of the groove.

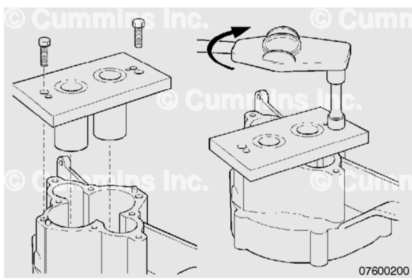
Note : If the bushing is allowed to block the oil groove, inadequate lubrication can result, causing excessive wear and premature malfunction of the bushing or gear shaft.



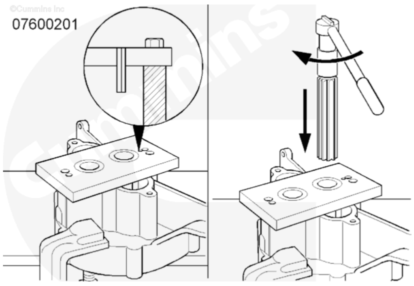
Mount fixture, Part Number 3823381, on the lubricating oil pump body. Use the two dowels and two capscrews from the pump.

Torque Value: 68 n•m [50 ft-lb]

Note : The fixture has a split pin that will extend into the gear pocket when mounted correctly.



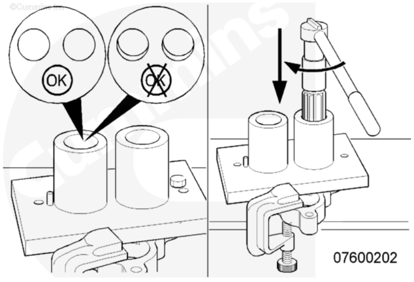
Secure the assembled body and fixture in a vise or clamp to keep it from moving while being reamed. Slip the reamer, Part Number 3823382, into a bushing in the fixture. Use a 1/2- in socket and ratchet to drive the reamer. Turn **clockwise** while applying moderate downward pressure on the reamer. When the reamer finishes cutting through the bushing, continue to turn **clockwise** while pulling up on the reamer to remove. Use cutting oil when reaming.



Note : A drill press or low rpm hand drill can be used to turn the reamer in place of the socket and ratchet, if available.

Repeat the reaming operation on the next hole.

Remove the fixture from the body and clean thoroughly. Mount the fixture over the lube pump dowels with the bushing UP on the cover. Carefully look through the fixture bushings and make sure that the lube pump shaft bushings are aligned.

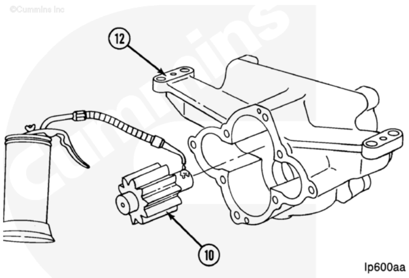


Repeat the reaming operation as in the last step.

Assemble

Two-Piece

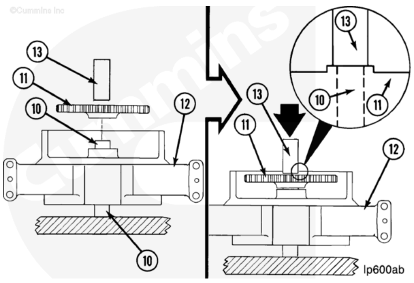
Use clean engine oil to lubricate the shaft.



Install the long end of the pumping drive gear and shaft assembly (10) through the bushing in the body (12) that is nearest to the top of the pump.

⚠ CAUTION ⚠

Do not damage the end of the shaft (10). The bushing will be damaged when the cover is installed.

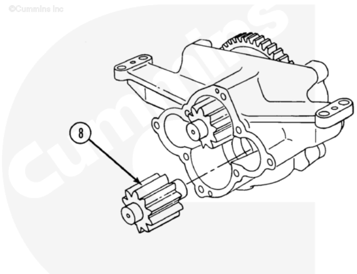


Use a hydraulic press and support the assembly on the pumping drive gear shaft as illustrated. Use a mandrel (13) that is larger than the shaft.

Push the main drive gear (11) on the shaft. Push the gear on the shaft until the hub of the gear is even with the end of the shaft.

Use clean engine oil to lubricate the lower bushing in the body.

Install the pumping idler gear and shaft (8).



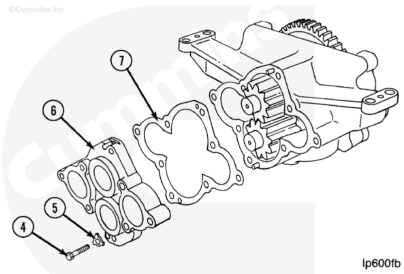
lp600fc

Use clean engine oil to lubricate the bushings in the cover.

Install the gasket (7) and the cover (6) on the dowel pins on the body.

Install the two new lockplates (5) and the two capscrews (4).

Torque Value: 40 n•m [30 ft-lb]

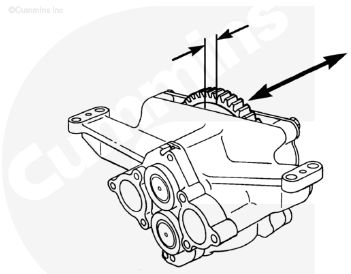


lp600fb

Move the gear by hand to measure the end play of the main drive gear.

Two-Piece Pump Drive Gear End Play		
mm		in
0.10	MIN	0.004
0.18	MAX	0.007

If the end play is **not** within specifications, the part **must** be rebuilt.

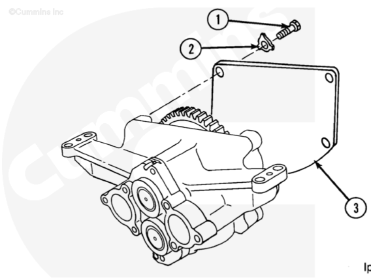


lp6geja

Install the cover (3), five new lockplates (2), and five capscrews (1).

Torque Value: 47 n•m [35 ft-lb]

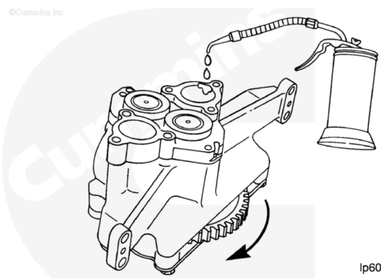
Bend the lockplates over the heads of the capscrews and over the edge of the cover.



lp600fa

Use clean engine oil to lubricate the pumping gears, while turning the main drive gear.

Cover the pump to prevent the entry of dirt until the pump is to be installed on the engine.

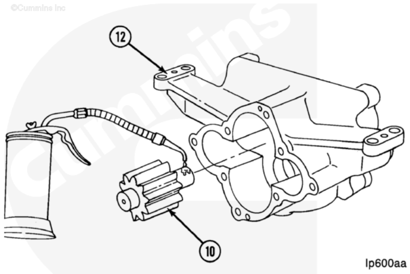


lp600wa

Three-Piece

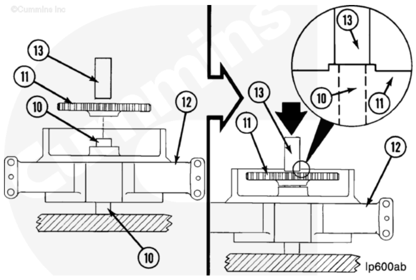
Use clean engine oil to lubricate the shaft.

Install the long end of the pumping drive gear and shaft assembly (10) through the bushing in the body (12) that is nearest to the top of the pump.



⚠ CAUTION ⚠

Do not damage the end of the shaft (10). The bushing will be damaged when the cover is installed.

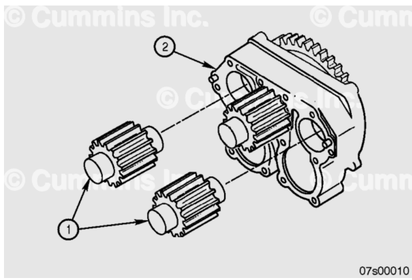


Use a hydraulic press and support the assembly on the pumping drive gear shaft as illustrated. Use a mandrel (13) that is larger than the shaft.

Push the main drive gear (11) on the shaft. Push the gear on the shaft until the hub of the gear is even with the end of the shaft.

Use clean engine oil to lubricate the remaining two bushings in the rear cover.

Install both pumping idler gears and shafts.



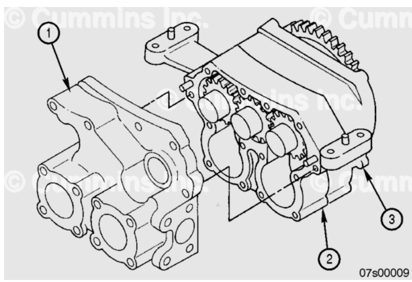
Install the pump body (2) onto the rear cover (3). Make sure the joint faces are clean and free from damage. Tap the body onto the dowels using a soft mallet.

Lubricate the bushings in the front cover (1).

Install the front cover (1). Make sure the joint faces are clean and free from damage. Tap the body onto the dowels using a soft mallet.

Install the four short and two long capscrews through the front cover and into the front body.

Tighten the capscrews.

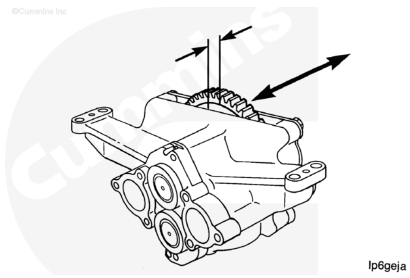


Torque Value: 95 n•m [70 ft-lb]

Move the gear by hand to measure the end play of the main drive gear.

Three-Piece Pump Drive Gear End Play		
mm		in
0.10	MIN	0.004
0.18	MAX	0.007

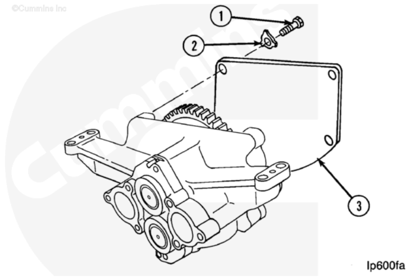
If the end play is **not** within specifications, the part **must** be rebuilt.



Install the cover (3), five new lockplates (2), and five capscrews (1).

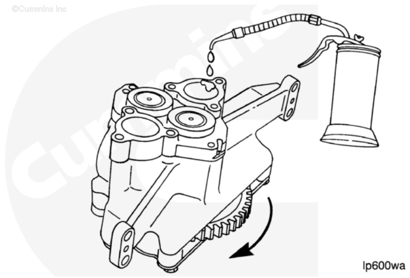
Torque Value: 47 n•m [35 ft-lb]

Bend the lockplates over the heads of the capscrews and over the edge of the cover.



Use clean engine oil to lubricate the pumping gears while turning the main drive gear.

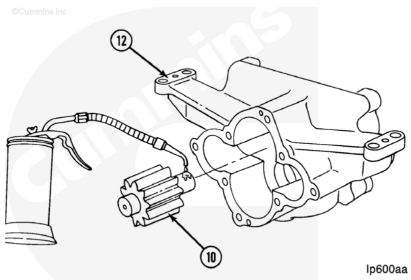
Cover the pump to prevent the entry of dirt until the pump is to be installed on the engine.



Slow-Speed

Use clean engine oil to lubricate the shaft.

Install the long end of the pumping drive gear and shaft assembly (10) through the bushing in the body (12) that is nearest to the top of the pump.

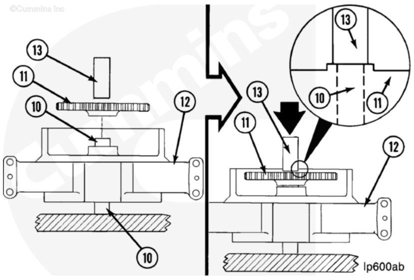


⚠ CAUTION ⚠

Do not damage the end of the shaft (10). The bushing will be damaged when the cover is installed.

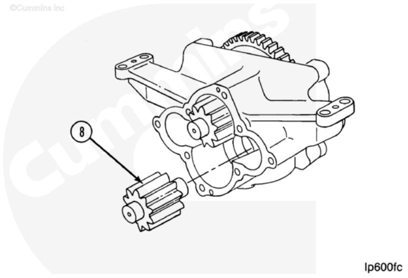
Use a hydraulic press. Support the assembly on the pumping drive gear shaft as illustrated. Use a mandrel (13) that is larger than the shaft.

Push the main drive gear (11) on the shaft. Push the gear on the shaft until the hub of the gear is even with the end of the shaft.



Use clean engine oil to lubricate the lower bushing in the body.

Install the pumping idler gear and shaft (8).

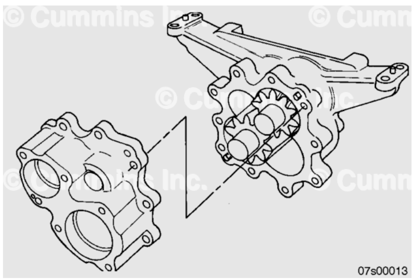


Use clean engine oil to lubricate the two bushings in the cover.

Install the cover. Tap the cover onto the dowels using a soft mallet.

Install and tighten the ten capscrews.

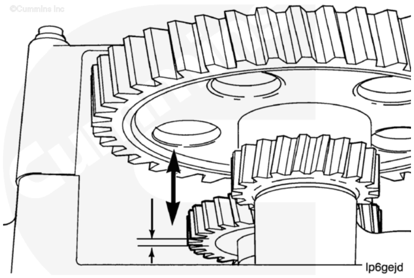
Torque Value: 95 n•m [70 ft-lb]



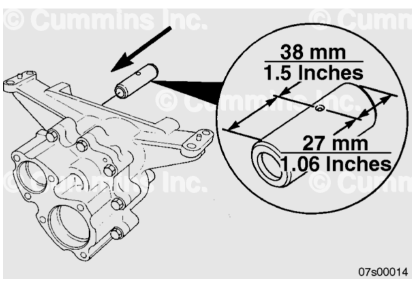
Move the gear by hand to measure the pumping gear end play.

Slow Speed Pump (Short Body and Long Body) Pumping Gear End Play		
mm		in
0.114	MIN	0.0045
0.216	MAX	0.0085

If the end play is **not** within specifications, the part **must** be rebuilt.



Install the idler shaft to the pump body, with the oil hole in the vertical position.

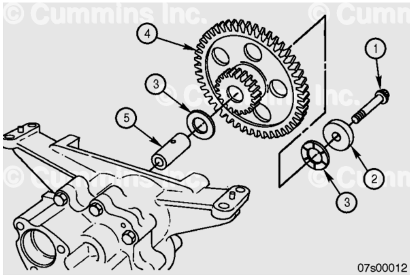


Use clean engine oil to lubricate the bore of the compound drive / idler gear (4) and the grooved faces of the thrust washers (3).

Install the idle shaft (5).

Install the compound gear (4), thrust washers (3), retainer (2) and capscrew (1) and tighten the capscrew to specification.

Torque Value: 169 n•m [125 ft-lb]

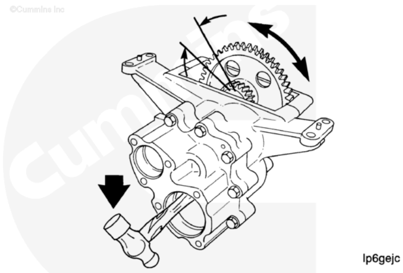


Note : Do not damage the pumping gears.

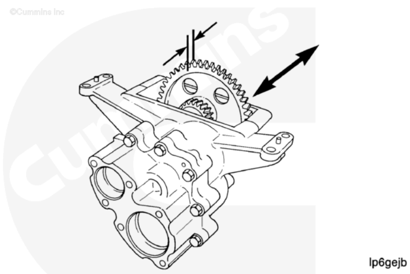
Use a piece of wood or the handle of a hammer to prevent the pumping gears from moving.

Move the gear by hand when checking the backlash.

Slow Speed Pump (Short Body and Long Body) Compound Gear to Drive Gear Backlash		
mm		in
0.15	MIN	0.006
0.25	MAX	0.010



Slow Speed Pump (Short Body and Long Body) Drive Gear End Play		
mm		in
0.20	MIN	0.008
0.81	MAX	0.032

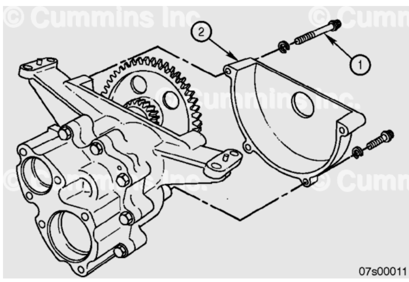


If the end play is **not** within specifications, the part **must** be rebuilt.

Install the drive gear cover (2) over the idler shaft retainer.

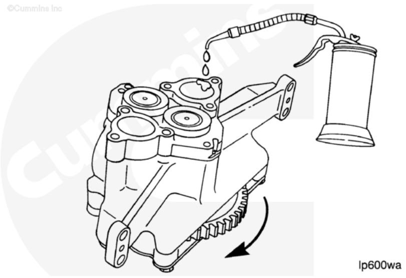
Install and tighten the four capscrews (1).

Torque Value: 47 n•m [35 ft-lb]



Use clean engine oil to lubricate the pumping gears while turning the main drive gear.

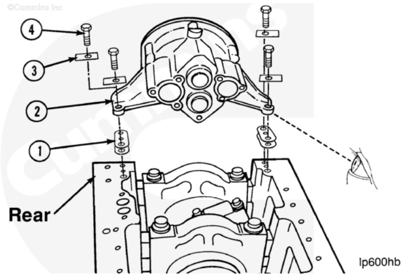
Cover the pump to prevent the entry of dirt until the pump is to be installed on the engine.



Install

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, make sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Note : Shown in this procedure is the K38. The procedure is the same for the K50, QSK38, KTA38GC, and QSK50 engines.

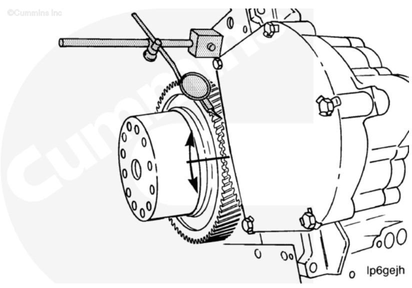
Check to make sure each mounting leg of the pump contains a dowel pin.

Install any shims (1) that were installed before the pump was removed. Install the pump (2) on the block. Install four new lockplates (3), the washers, and the capscrews (4).

Torque Value: 95 n•m [70 ft-lb]

Bend the lockplates over the heads of the capscrews and the feet of the pump.

Measure the lubricating oil pump drive gear backlash.



Note : The top of the engine must be positioned UP to measure the backlash of the lubricating oil pump drive gear.

Use a dial indicator to measure the pump drive gear-to-rear crankshaft gear backlash.

Lubricating Pump-to-Rear Crankshaft Gear Backlash		
mm		in
0.08	MIN	0.003

Lubricating Pump-to-Rear Crankshaft Gear Backlash		
mm		in
0.30	MAX	0.012

If the backlash is less than specifications, remove the pump and install shims of equal thickness between each of the mounting legs and the cylinder block. Shims are available in 0.13 mm [0.005 in], Part Number 3026312, and 0.25 mm [0.010 in], Part Number 3026313.

Install the pump. Use the previous procedure. Measure the backlash again.

If the backlash is greater than specifications, remove any shims and check the backlash again. If no shims were used, the pump drive gear or the rear crankshaft gear is worn and **must** be replaced.

Finishing Steps

⚠ **WARNING** ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, make sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Install the lubricating oil suction tube (block-mounted). Refer to Procedure 007-035 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-035-tr.html)
- Install the lubricating oil pan adapter. Refer to Procedure 007-027 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-027-tr.html)
- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-025-tr.html)
- Install the full flow filters. Refer to Procedure 007-013 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-013-tr.html)

- Fill the engine with lubricating oil. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-037-tr.html)
- Connect the batteries. Refer to the equipment manufacturer service information.
- Operate the engine and check for leaks.

Last Modified: 29-Mar-2016